Day 67 — TNPSC Group 2A Study Material

Part A – General Studies: Current Affairs in Geography (3): National Parks & Cyclones

READING MATERIAL

Covers India's most recently designated national park, current national park/wildlife sanctuary totals, and the 2025 North Indian Ocean cyclone season — including the system for naming cyclones.

Q: What is India's newest (107th) National Park, and when was it designated?
A: Similipal National Park — officially designated India's 107th National Park in April 2025.

Q: As of 2026, how many National Parks and Wildlife Sanctuaries does India have in total?
A: 107 National Parks and 573 Wildlife Sanctuaries.

Q: Which is India's largest National Park by area, and where is it located?
A: Hemis National Park — located in eastern Ladakh, spanning approximately 4,400 square kilometres.

Q: What made the 2025 North Indian Ocean cyclone season particularly notable, and which specific cyclone caused major loss of life?
A: It was the COSTLIEST season on record and the DEADLIEST since 2008. Cyclone Ditwah, which meandered near Sri Lanka in November 2025, killed 647 people in that country.

Q: How are North Indian Ocean cyclones named, and how many countries/names are involved in the latest list?
A: The India Meteorological Department (IMD) maintains a rotating list of names contributed by 13 countries (India, Sri Lanka, Maldives, Myanmar, Oman, Thailand, Qatar, UAE, Saudi Arabia, Yemen, Pakistan, Bangladesh, Iran) — the latest list comprises 169 names, with India contributing 13 names including Gati, Tej, Aag, Neer, Vyom, and Jaladhi.

MOCK TEST (WITH ANSWERS)

1. Which National Park became India's 107th, designated in April 2025?

A. Hemis National Park
B. Similipal National Park ✓
C. Kaziranga National Park
D. Gir National Park

Explanation: Similipal National Park. [medium, web-research]

2. As of 2026, how many National Parks does India have in total?

A. 95
B. 101
C. 107 ✓
D. 115

Explanation: 107. [medium, web-research]

3. As of 2026, how many Wildlife Sanctuaries does India have?

A. 350
B. 450
C. 573 ✓
D. 650

Explanation: 573. [hard, web-research]

4. Which is India's largest National Park by area, located in Ladakh?

A. Hemis National Park ✓
B. Similipal National Park
C. Corbett National Park
D. Sundarbans National Park

Explanation: Hemis National Park. [medium, web-research]

5. Cyclone Ditwah (November 2025) caused major loss of life in which country?

A. India

B. Sri Lanka ✓

C. Bangladesh

D. Myanmar

Explanation: Sri Lanka. [medium, web-research]

6. How many countries contribute names to the North Indian Ocean cyclone-naming list maintained by the IMD?

A. 8

B. 10

C. 13 ✓

D. 20

Explanation: 13 countries. [hard, web-research]

Part B – Aptitude & Mental Ability: Highest Common Factor (HCF)

READING MATERIAL

*HCF is 1 of 10 named Aptitude sub-skills. Real exam evidence (4 confirmed hits across 2022/2024/2025) shows TNPSC favors HCF *application* problems — measuring/tiling/container word problems and the HCF-LCM relationship — over plain 'find the HCF of these two numbers' questions, so the reading below leads with those application shapes.*

Q: Real exam (2025): Two numbers are in the ratio 15:11. If their HCF is 13, find the numbers.

A: When two numbers are in a given ratio, write them as 15k and 11k. Since HCF(15,11)=1 (the ratio is already in lowest terms), k IS the HCF — so k=13. The numbers are 15×13=195 and 11×13=143.

Q: Real exam (2025): What is the least number of square tiles needed to pave the ceiling of a room 15m 17cm long and 9m 2cm broad?

A: Convert to one unit: 1517 cm × 902 cm. The LARGEST square tile that exactly fits (no cutting) has a side equal to the HCF of the two dimensions. HCF(1517, 902) = 41 cm (via the Euclidean algorithm: 1517=1×902+615; 902=1×615+287; 615=2×287+41; 287=7×41+0). Tile area = 41×41 = 1,681 cm^2. Room area = 1517×902 = 1,368,334 cm^2. Number of tiles = 1,368,334 ÷ 1,681 = 814.

Q: Real exam (2024): If the HCF of x^3-8y^3 and $x^2-k^2y^2$ is x+ky, find k.

A: Factor x^3-8y^3 using $a^3-b^3=(a-b)(a^2+ab+b^2)$ with a=x, b=2y: $x^3-8y^3 = (x-2y)(x^2+2xy+4y^2)$. Factor $x^2-k^2y^2$ using $a^2-b^2=(a-b)(a+b)$: $x^2-k^2y^2 = (x-ky)(x+ky)$. For 'x+ky' to be the common factor (HCF) of both, it must match one of the first expression's factors. (x−2y) is the only linear factor available there, so x+ky must equal x−2y, giving k=−2.

Q: Real exam (2022): What is the greatest possible volume of a vessel that can exactly measure the milk in cans of 80, 100, and 120 litres (each can measured to full capacity only)?

A: This is asking for the HCF of 80, 100, and 120. HCF(80,100)=20, then HCF(20,120)=20. So the answer is 20 litres — the largest vessel that can measure each can's full volume an exact whole number of times.

Q: Real exam (2022): The HCF and LCM of two numbers are 16 and 240 respectively. If one number is 48, find the other.

A: Use the relationship: HCF × LCM = product of the two numbers. 16 × 240 = 3,840. Other number = 3,840 ÷ 48 = 80.

Q: Method — finding HCF via the Euclidean algorithm (the fast way for large numbers): What is the HCF of 1517 and 902?

A: Repeatedly divide the larger by the smaller and replace the larger with the remainder, until the remainder is 0 — the last nonzero remainder is the HCF. 1517=1×902+615 → 902=1×615+287 → 615=2×287+41 → 287=7×41+0. HCF=41. This is the same calculation used in r02 above — recognize it as the standard fast method rather than listing all factors of both numbers.

MOCK TEST (WITH ANSWERS)

1. Two numbers are in the ratio 21:14. If their HCF is 12, find the larger number.

A. 36 ✓

B. 24

C. 252

D. 168

Explanation: Watch out: 21 and 14 are NOT coprime (gcd=7), so you can't directly multiply them by the HCF. Reduce the ratio first: 21:14 = 3:2 (coprime). Now apply the HCF: larger = 3×12=36, smaller = 2×12=24. [hard, authored]

2. Find the largest square tile that can exactly pave a floor measuring 3m 8cm by 4m 20cm (no cutting).

A. 28 cm ✓

B. 14 cm

C. 56 cm

D. 84 cm

Explanation: Convert to cm: 308 and 420. HCF(308,420): 420=1×308+112; 308=2×112+84; 112=1×84+28; 84=3×28+0. HCF=28 cm. [hard, authored]

3. If the HCF of a^2-4b^2 and $a^2+ab-6b^2$ is a−2b, this is consistent with which factorization of the second expression?

A. (a+3b)(a−2b) ✓

B. (a−3b)(a+2b)

C. (a+3b)(a+2b)

D. (a−3b)(a−2b)

Explanation: $a^2-4b^2=(a-2b)(a+2b)$. The second expression factors as $(a+3b)(a-2b) = a^2+ab-6b^2$ — sharing the common factor (a−2b), so that's the HCF. [hard, authored]

4. What is the greatest volume that can exactly measure the contents of drums holding 72, 96, and 120 litres?

A. 12 litres

B. 24 litres ✓

C. 8 litres

D. 6 litres

Explanation: HCF(72,96)=24, then HCF(24,120)=24. Answer: 24 litres. [medium, authored]

5. The HCF and LCM of two numbers are 12 and 180 respectively. If one number is 36, find the other.

A. 48

B. 54

C. 60 ✓

D. 72

Explanation: HCF×LCM = 12×180 = 2,160. Other number = 2,160 ÷ 36 = 60. [medium, authored]

6. Using the Euclidean algorithm, find the HCF of 1,073 and 925.

A. 37 ✓

B. 29

C. 31

D. 43

Explanation: 1073=1×925+148; 925=6×148+37; 148=4×37+0. HCF=37. [medium, authored]

7. Two numbers are in the ratio 8:9 and their HCF is 7. Find their sum.

A. 105

B. 119 ✓

C. 126

D. 112

Explanation: Numbers: 8×7=56 and 9×7=63. Sum = 56+63 = 119. [medium, authored]

Part C – General English: Writing Skills: Jumbled Sentences and Finding the Right Order

READING MATERIAL

Covers the skill of rearranging 4 jumbled sentences into a logical, coherent paragraph — a pure reasoning/sequencing skill, tested via complete worked examples.

Q: What general strategy helps identify the FIRST sentence in a jumbled-sentence question?

A: The first sentence is usually an independent, general/introductory statement — setting the scene, stating a general fact, or starting an incident (often starting with a noun, 'most', or 'once') — rather than a sentence that depends on something mentioned earlier (like 'it', 'he', 'then', 'finally').

Q: What clues help link the middle and final sentences together correctly?

A: Connector/transition words (e.g., 'but', 'when', 'then', 'therefore', 'however', 'finally') signal how one sentence follows from another. Pronouns ('it', 'he', 'they') must refer back to something already introduced in an earlier sentence — a sentence using 'it' typically cannot come first if nothing has been introduced yet.

Q: Worked example: "1. The train was late. 2. We reached the station on time. 3. We waited for one hour. 4. Finally, it arrived." What is the correct order, and why?

A: 2-3-1-4: We reached the station ON TIME (2, sets the scene) → we WAITED for one hour (3, because something was wrong) → THE TRAIN WAS LATE (1, explains why we waited) → FINALLY, it arrived (4, concluding with 'finally', a clear ending signal).

Q: Worked example: "1. I went to the library. 2. I borrowed a book. 3. I returned it after a week. 4. I enjoyed reading it." What is the correct order, and why?

A: 1-2-4-3: I went to the library (1, introductory action) → I borrowed a book (2, the next logical step) → I enjoyed reading it (4, what happened with the book) → I returned it after a week (3, the concluding action, AFTER reading it — note 'after a week' signals this comes near the end).

MOCK TEST (WITH ANSWERS)

1. Arrange in the correct order: "1. The train was late. 2. We reached the station on time. 3. We waited for one hour. 4. Finally, it arrived."

A. 2-1-3-4
B. 1-3-2-4
C. 1-2-3-4
D. 2-3-1-4 ✓

Explanation: 2-3-1-4. [medium, web-research]

2. Arrange in the correct order: "1. Honesty is a virtue. 2. It builds character. 3. Everyone respects an honest person. 4. It is always rewarded."

A. 1-2-3-4 ✓
B. 1-2-4-3
C. 1-3-2-4
D. 3-1-2-4

Explanation: 1-2-3-4. [medium, web-research]

3. Arrange in the correct order: "1. The dog barked loudly. 2. It saw a stranger. 3. The owner calmed it down. 4. He gave it a treat."

A. 2-1-3-4
B. 1-2-3-4 ✓
C. 1-3-2-4
D. 3-1-4-2

Explanation: 1-2-3-4. [medium, web-research]

4. Arrange in the correct order: "1. I went to the library. 2. I borrowed a book. 3. I returned it after a week. 4. I enjoyed reading it."

A. 1-2-3-4
B. 1-2-4-3 ✓
C. 1-3-2-4
D. 1-4-2-3

Explanation: 1-2-4-3. [hard, web-research]

5. Arrange in the correct order: "1. The farmer ploughed the field. 2. He sowed the seeds. 3. Then he watered the field. 4. After some time, crops grew."

A. 1-2-3-4 ✓

B. 2-1-3-4

C. 3-2-1-4

D. 4-3-2-1

Explanation: 1-2-3-4. [easy, web-research]

6. Arrange in the correct order: "1. The children played in the park. 2. They climbed the trees. 3. They enjoyed the swings. 4. They laughed happily."

A. 1-2-3-4 ✓

B. 1-3-2-4

C. 1-2-4-3

D. 1-4-2-3

Explanation: 1-2-3-4. [easy, web-research]